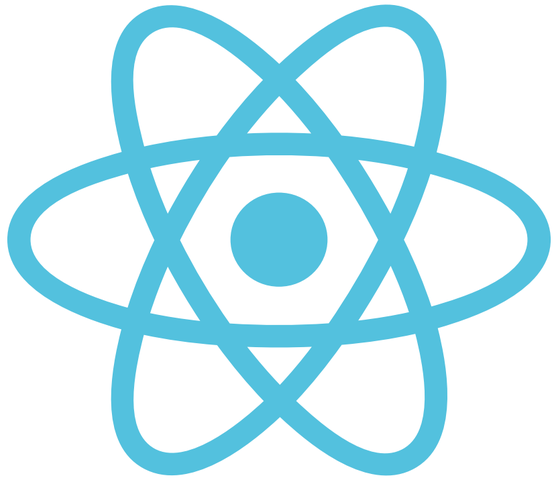
By: Mahmoud Abdelaziz

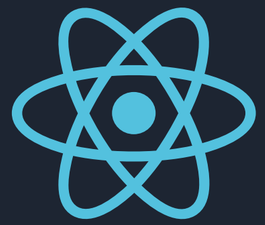
REACT JS



CONTENT

- JSX
- jsx syntax limitations
- Class vs Fn. Component
- State & Props & Children Props
- Life Cycle Methods
- Events
- Form
- Routing

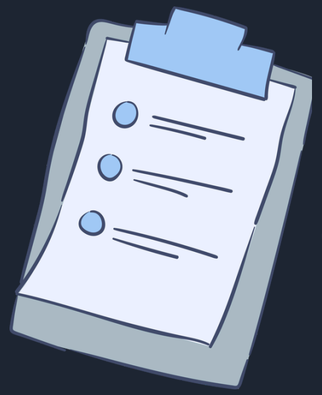
- Styling
- Pure Components
- React hooks



- Context API
- React.Lazy
- Lifting State up
- uuid , conditional Render
- Call Api

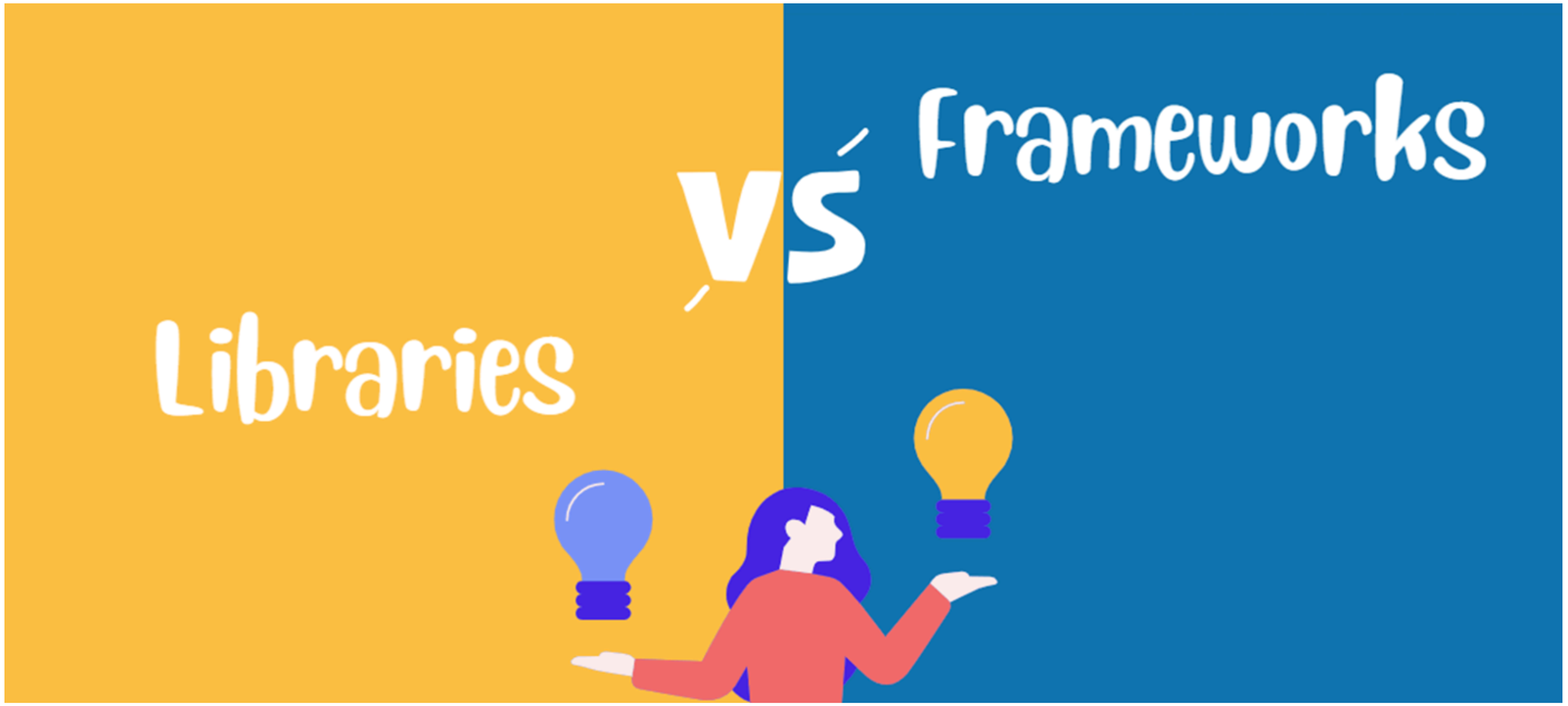
DAY 1

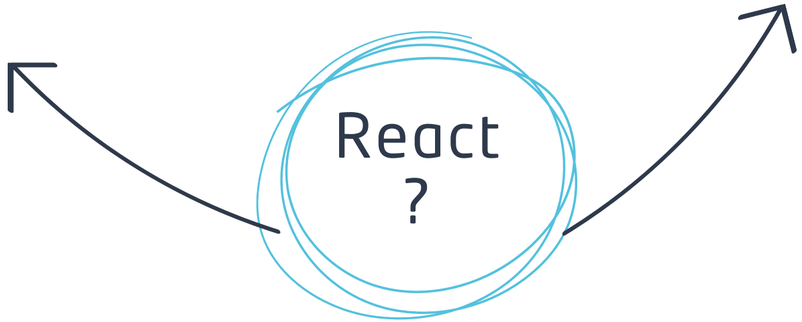
- Introduction
  - Framework Vs Library
  - Tools Behind
  - Why React
  - MPA Vs SPA
- How React Engine Works
- Create React Project
- Components
  - Class Comp
  - Function Comp
  - State And Props



Agenda

Framework vs Library





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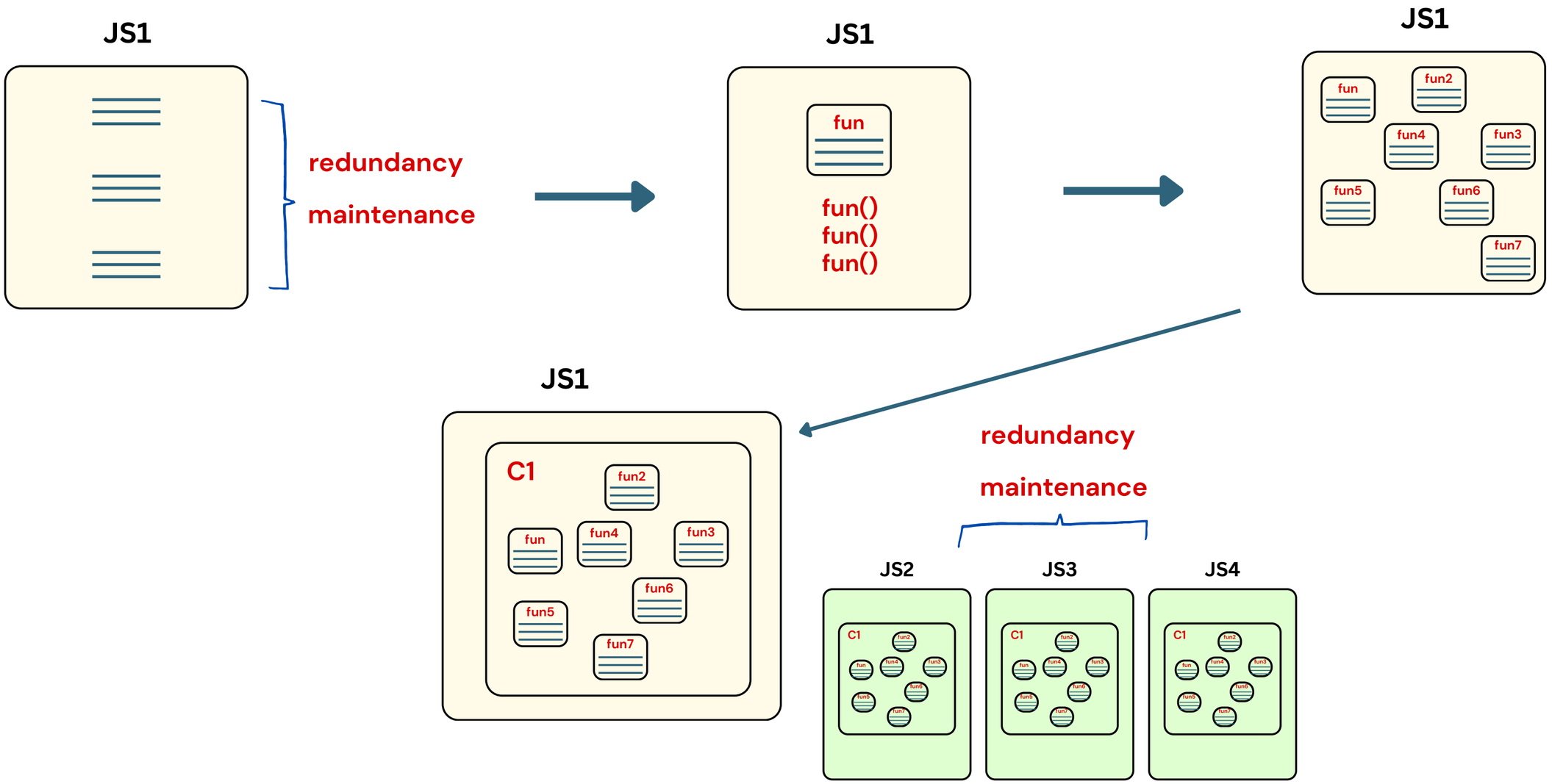

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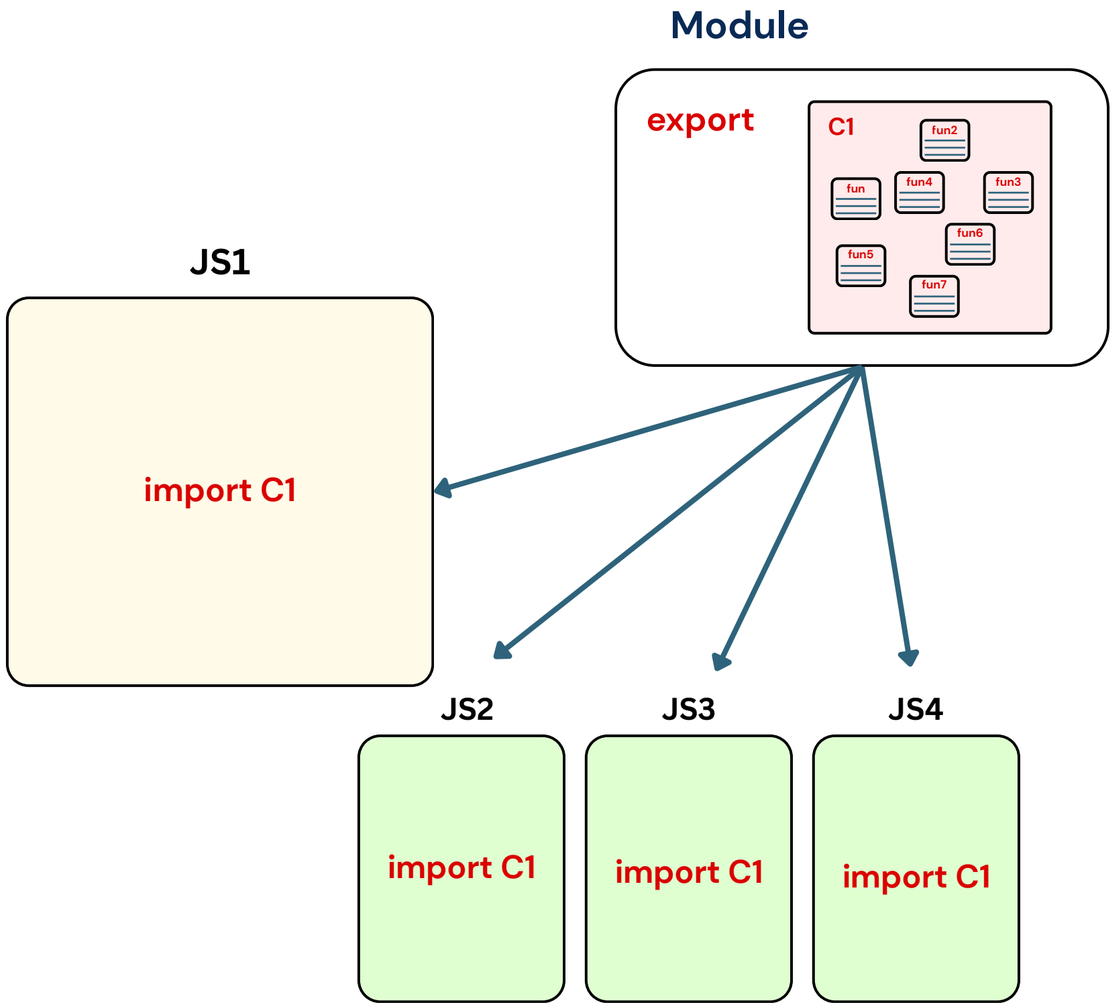
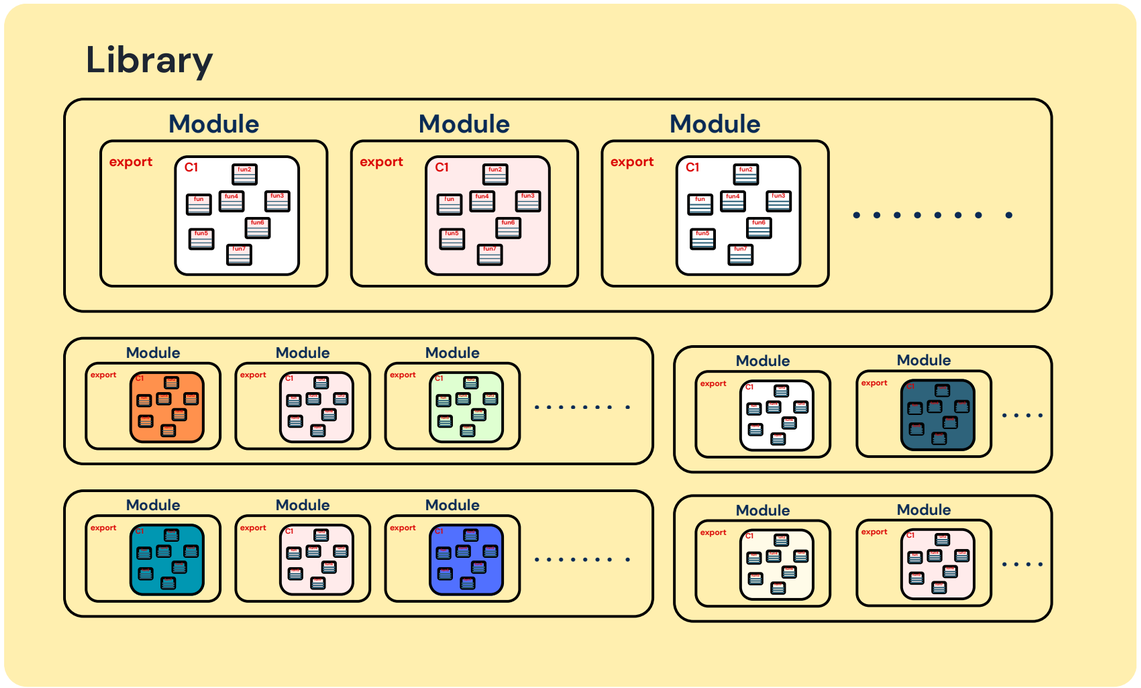
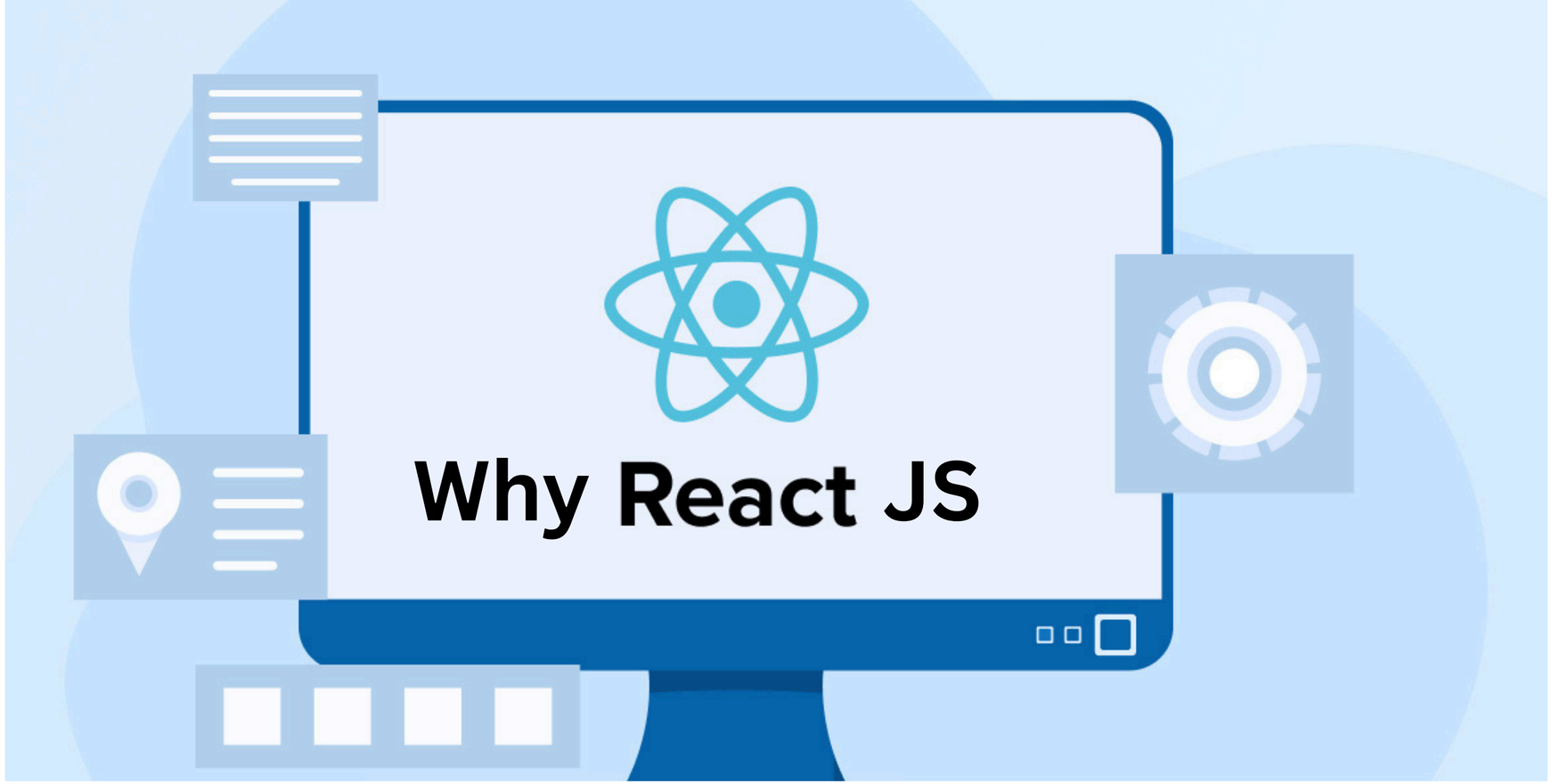


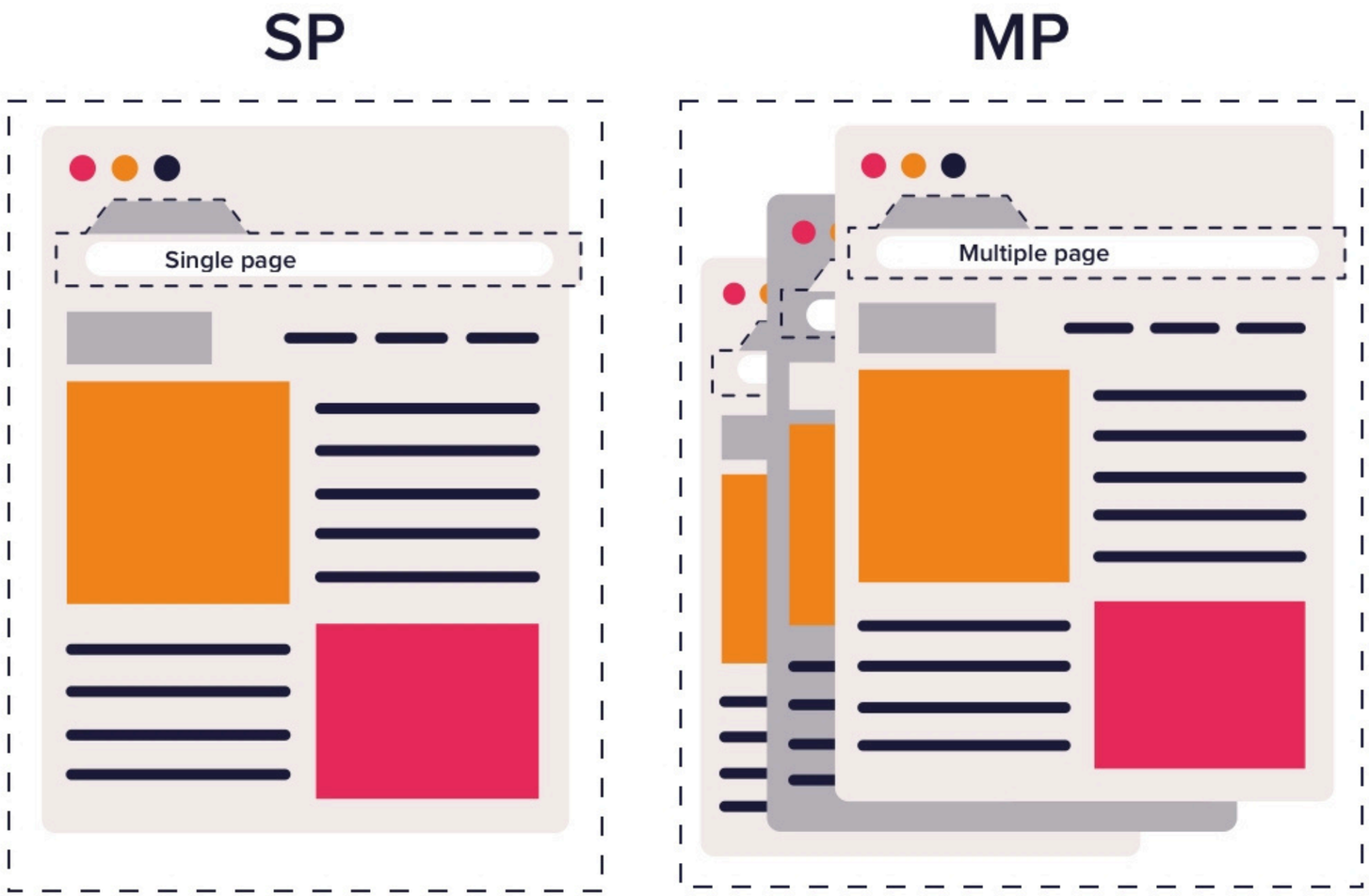
Framework



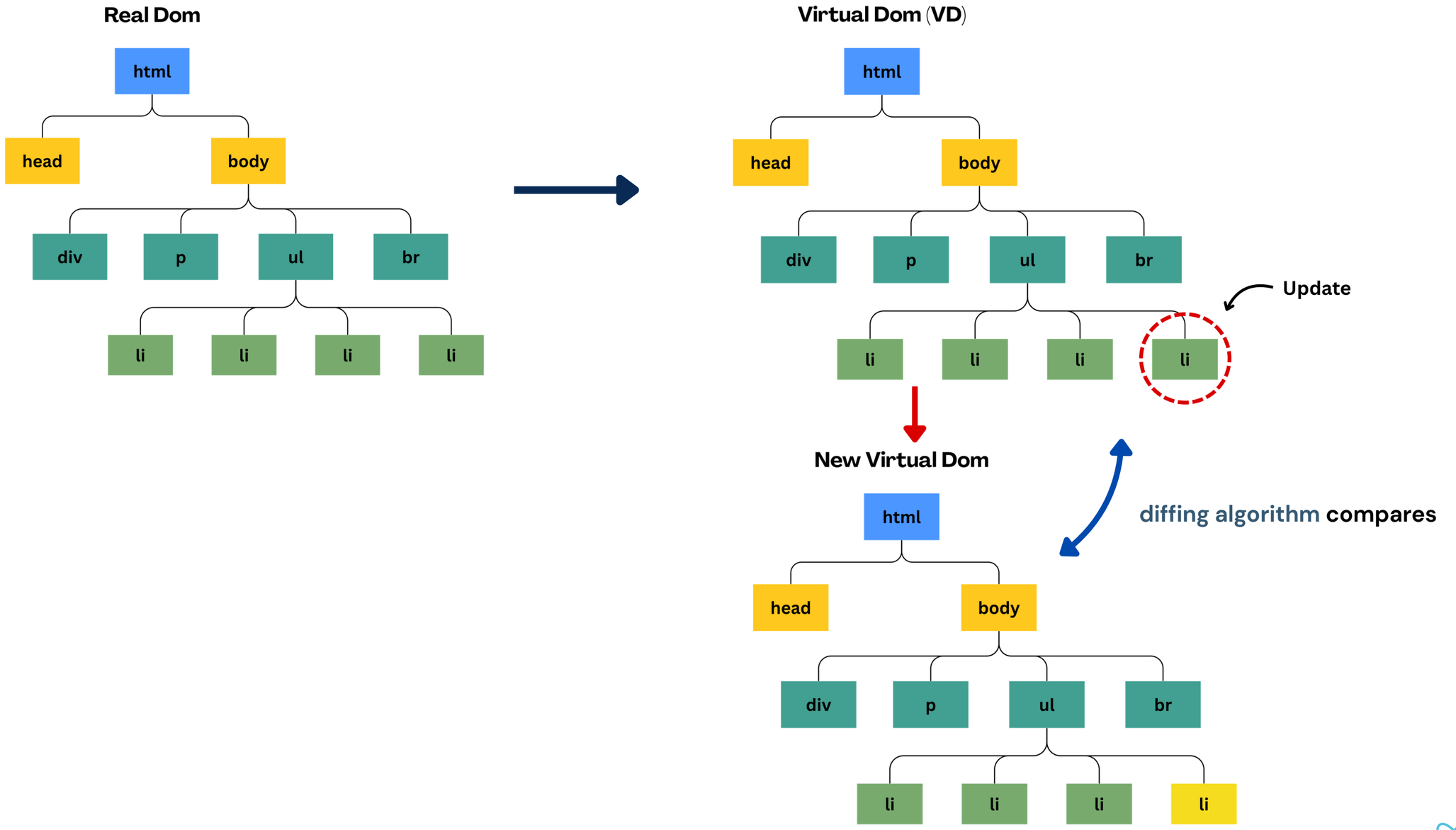


React.js was created by **Jordan Walke**, a software engineer at Facebook.
He developed React in **2011**.
React was first used in Facebook's news feed and **2012** on Instagram
before being open-sourced in **2013**. Since then, it has grown to become one of the most popular libraries for building web applications.

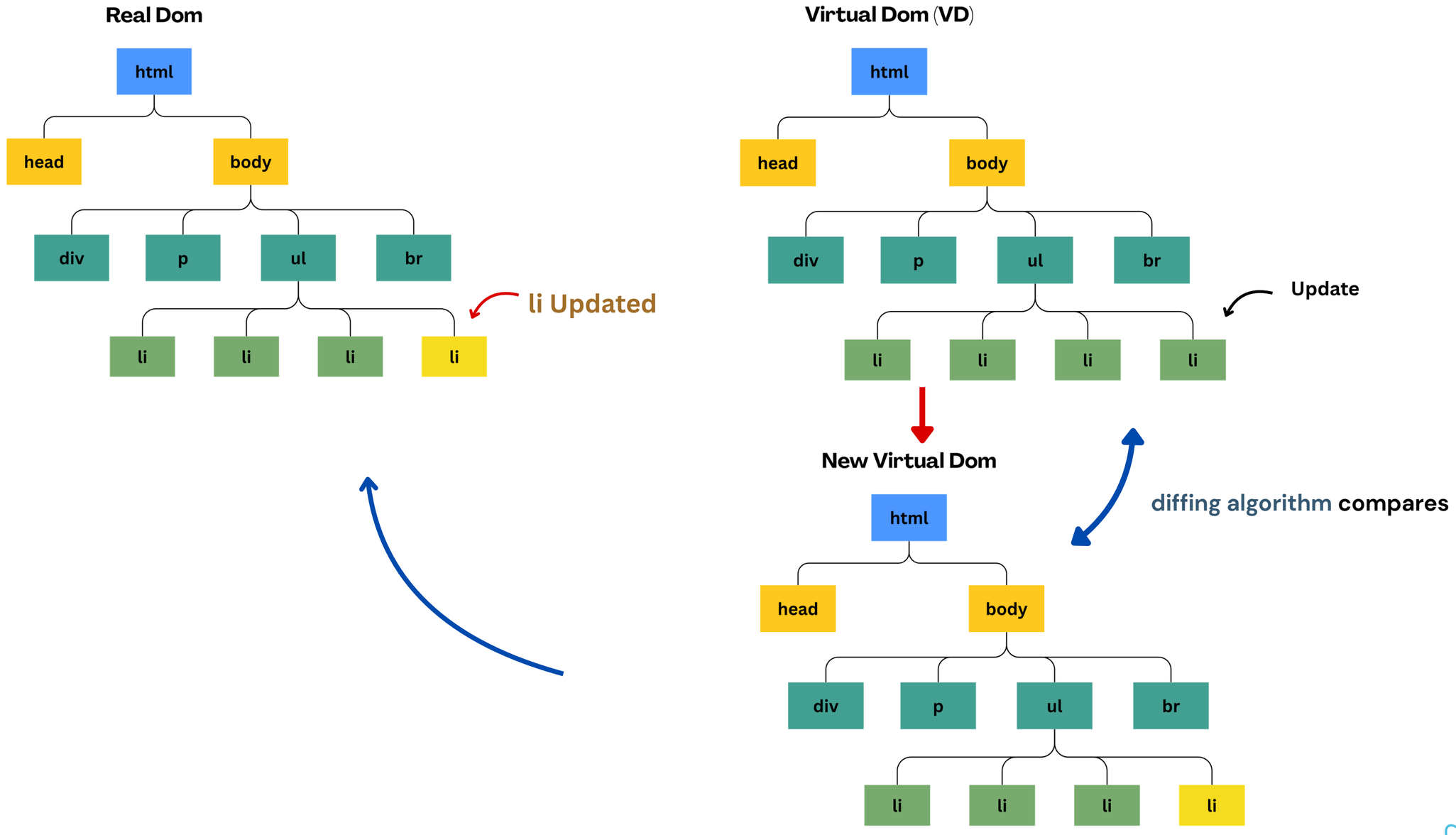
MPA Vs SPA



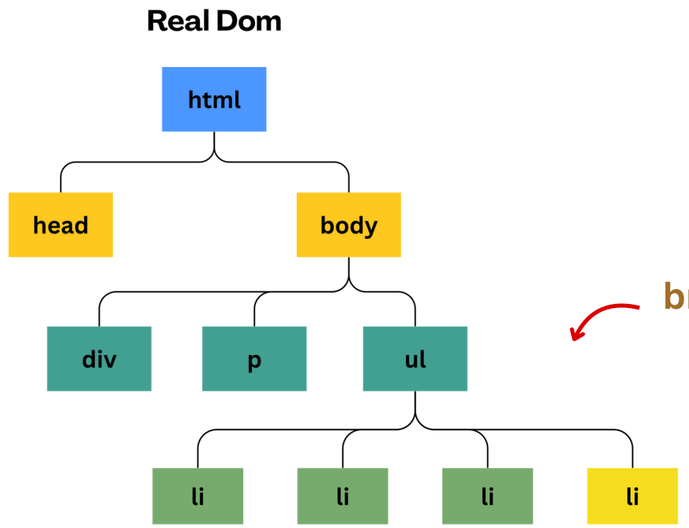
How React Works



How React Works



How React Works



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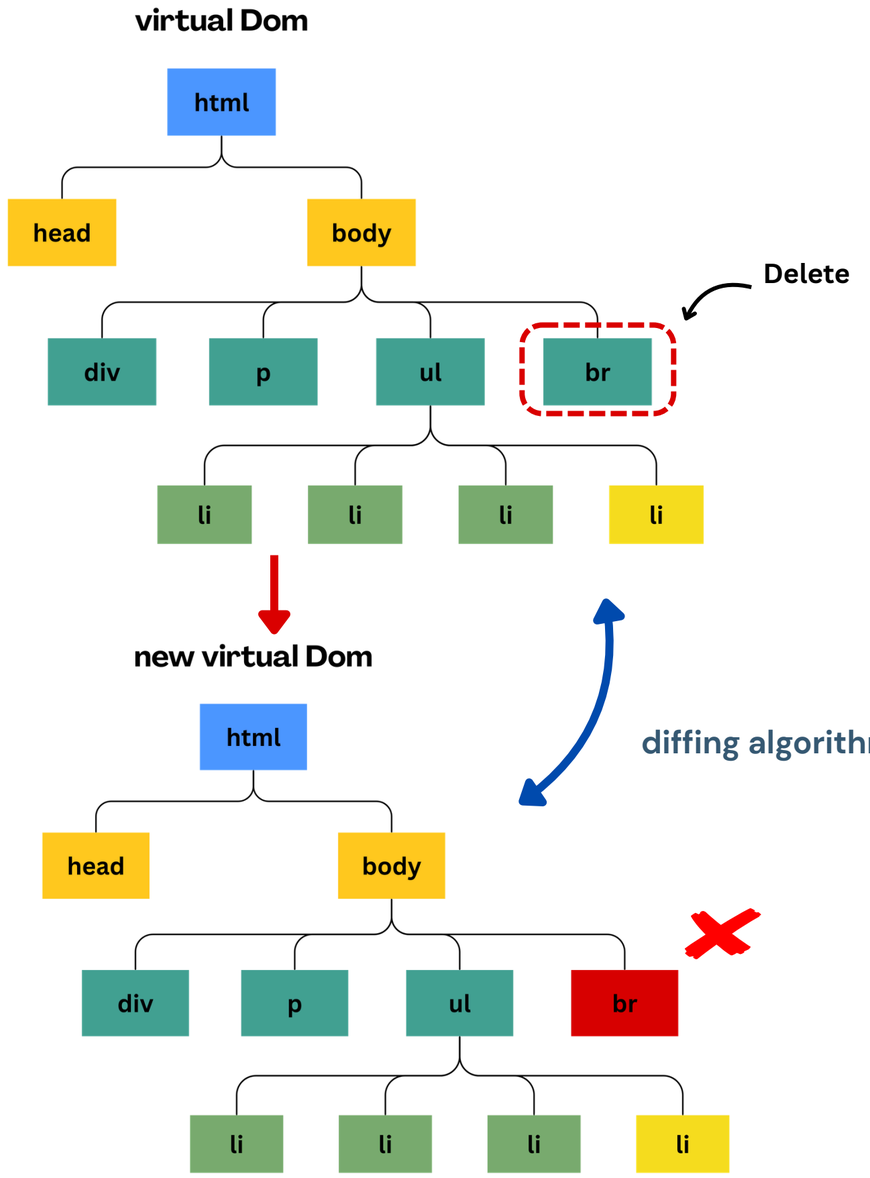


diffing algorithm compares

- Reconciliation Process
- Diffing Algorithm
- Fiber Engine

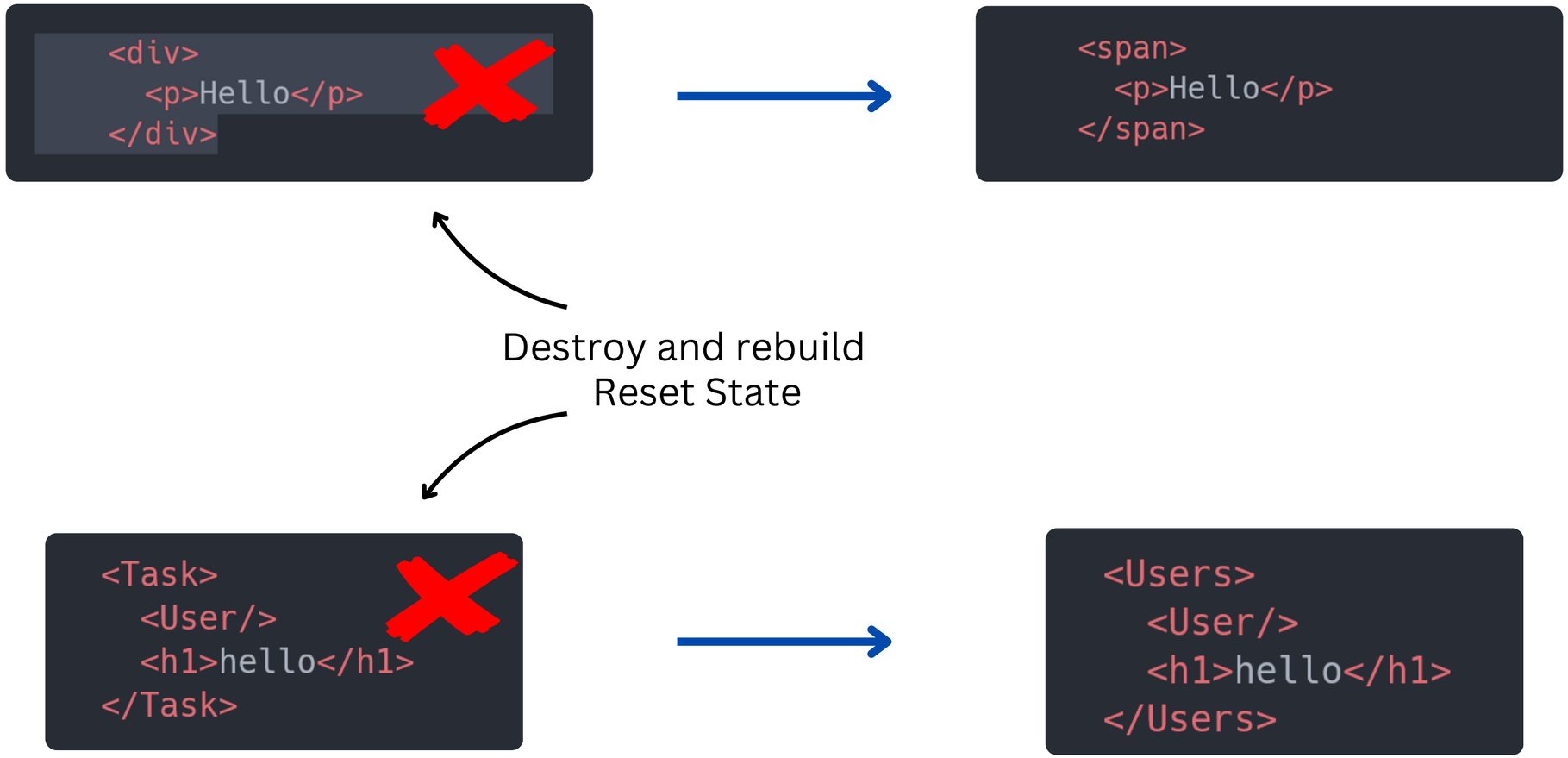
First Rule 👉 “Don’t Mutate”

Shallow comparison O(n) ✅

Deep comparison O(n)3

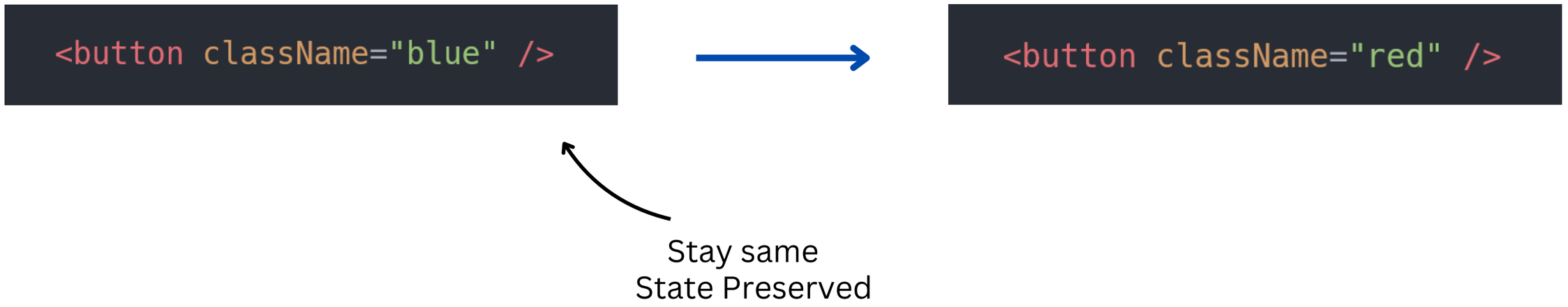
Diffing rules

1. Different Elements Same Position



Diffing rules

2. Same Elements Same Position



Component - Component instance - React Element

Component 🤔

A Component is a reusable function or class that defines how a part of the UI should look and behave.

```jsx
function User(){
  return(
    <>
      <div className="content">
        <h1>this is a user component</h1>
        <span>hello</span>
      </div>
    </>
  )
}
export default User;
```

Component Instance

A Component Instance is a specific execution of a component with its own state, props, and lifecycle.

```jsx
<User></User>
```

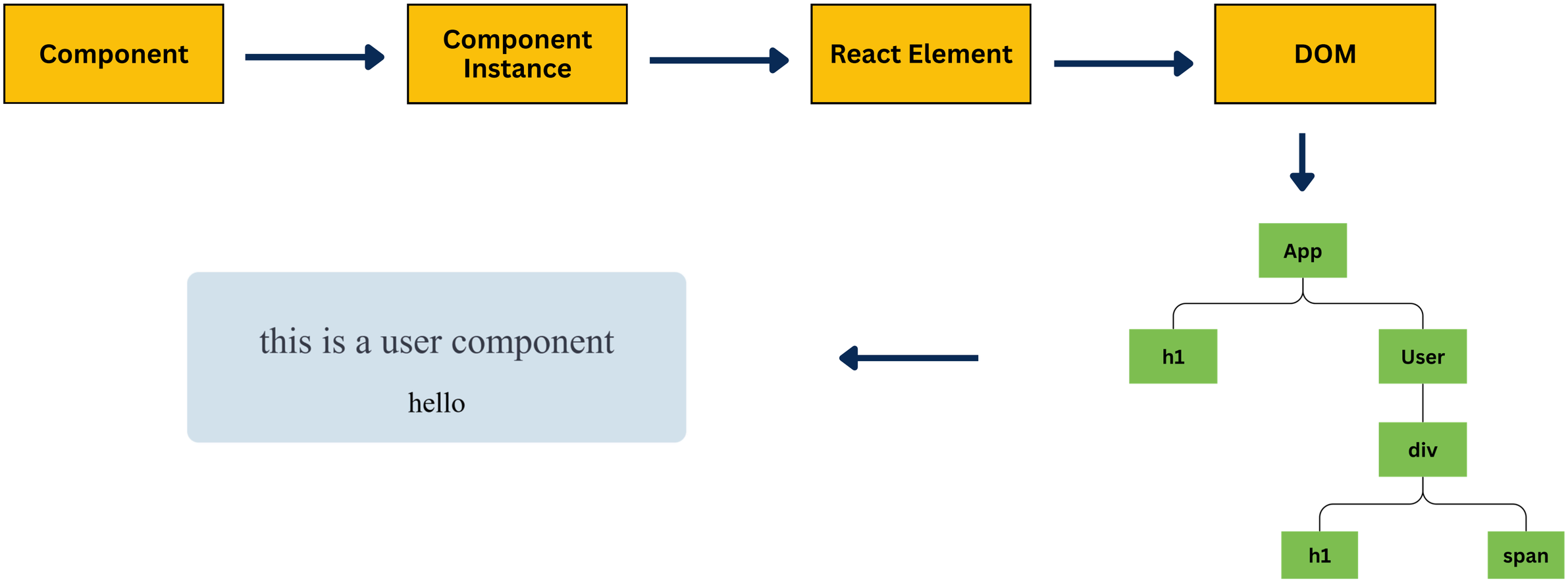
React Element (VD)

A React Element is a plain JavaScript object that describes what should appear on the U

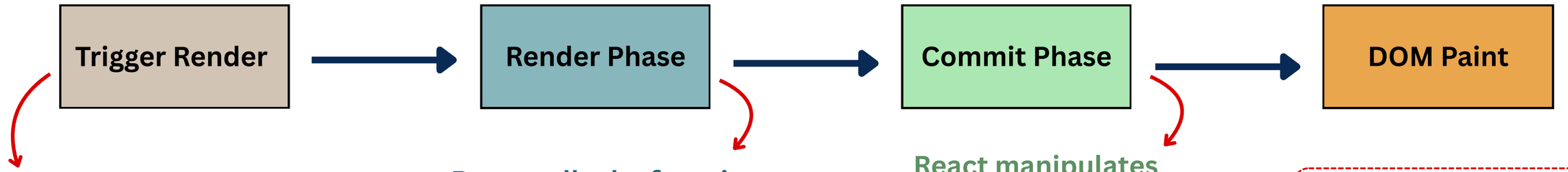
```jsx
React.createElement(
    React.Fragment,
    null,
    React.createElement(
      "div",
      { className: "content" },
      React.createElement(
        "h1",
        null,
        "this is a user component"
      ),
      React.createElement(
        "span",
        null,
        "hello"
      )
    )
  );
```

```jsx
{
  $$typeof: Symbol(react.transitional.element),
  key: null,
  props: {},
  type: function User() {},
  _owner: {tag: 0,key: null,stateNode: null,elementType: function User() {},
  _store: {validated: 0},
  ref: null,
  _debugInfo: null,
  _debugStack: "Error: react-stack-top-frame",
  _debugTask: {
    run: function () {}
  }
};
```

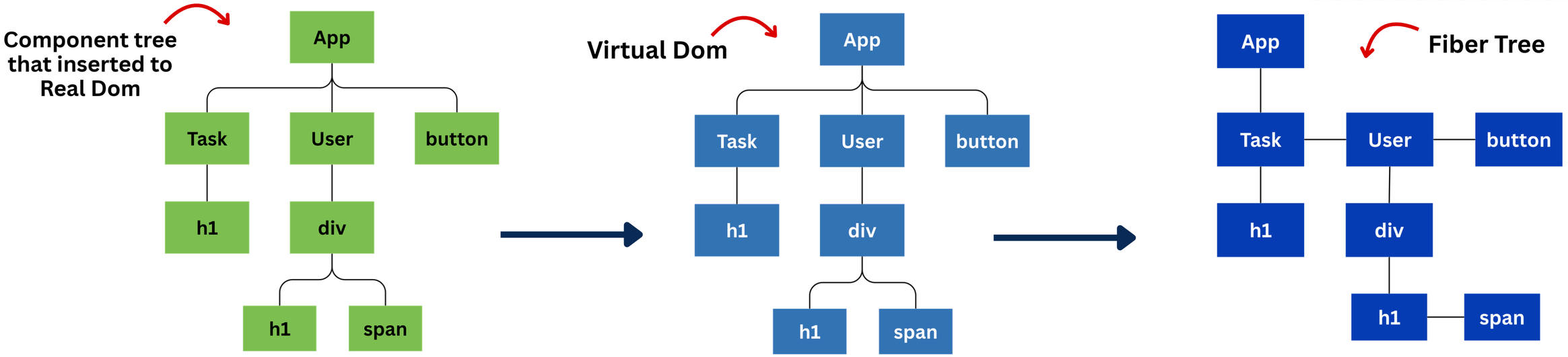
Component - Component instance - React Element



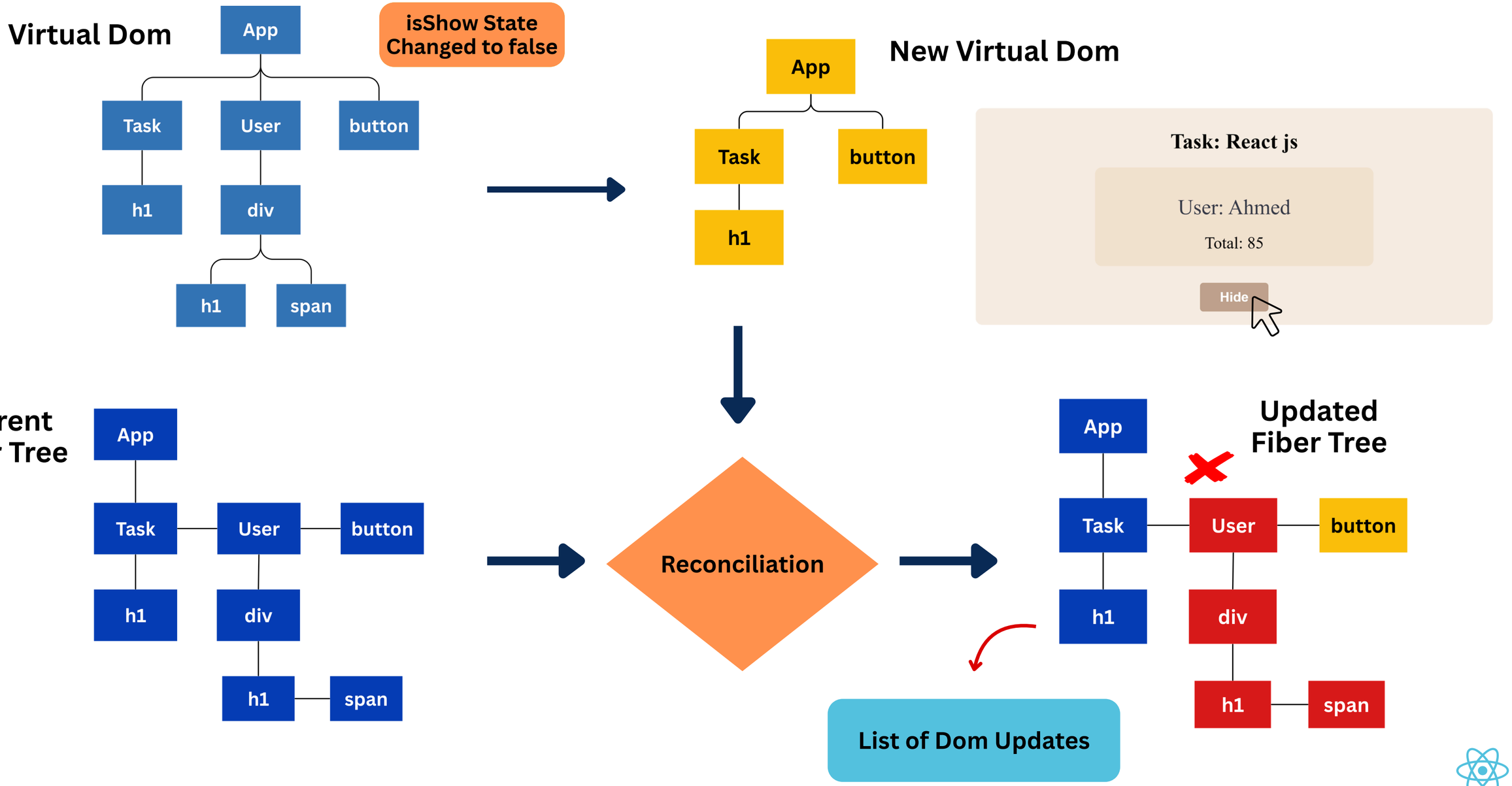
From Component To DOM



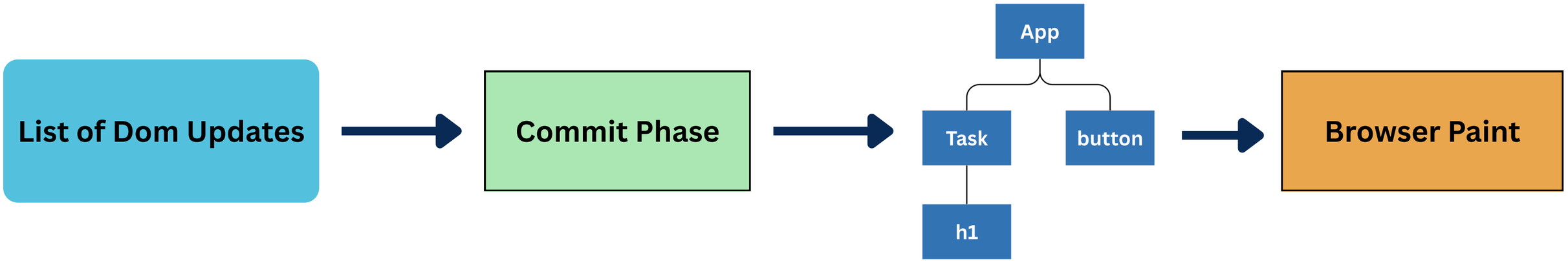
The Fiber Tree:
is React's internal data structure that represents the component hierarchy and stores all the information needed to efficiently render, update, and manage the UI.



Re-Render With Reconciliation Process



Re-Render With Reconciliation Process

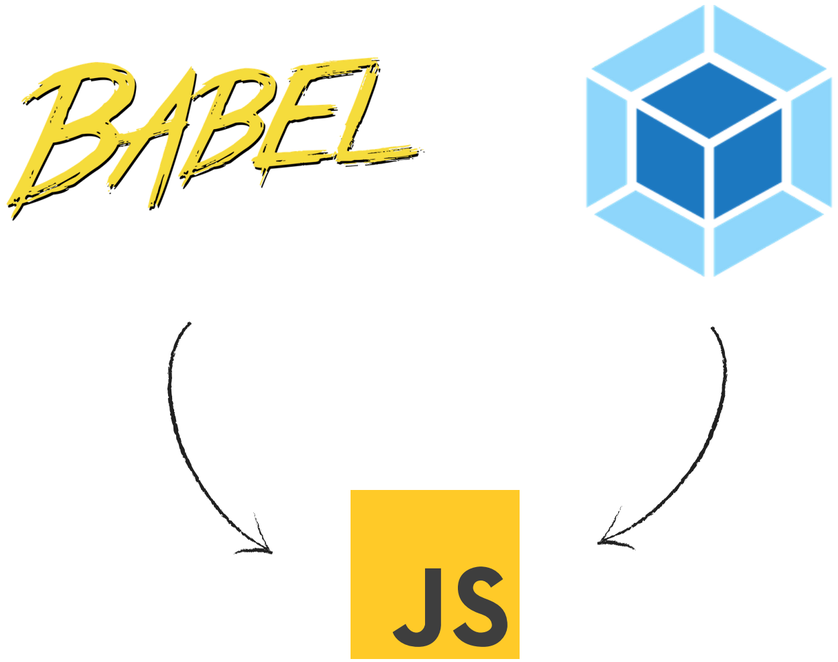


Task: React js

Show

Tools Behind

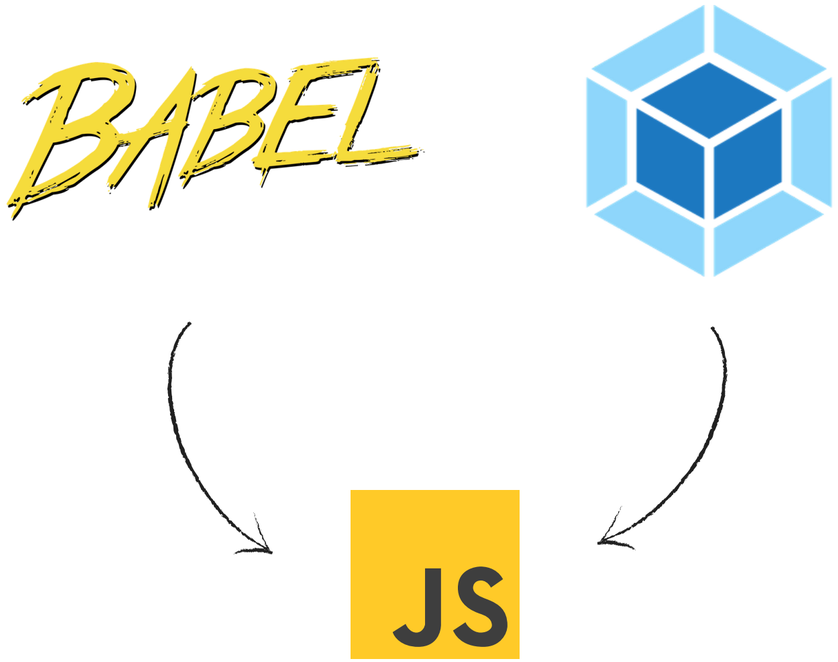
With Create-React-App Library



With Vite

Tools Behind

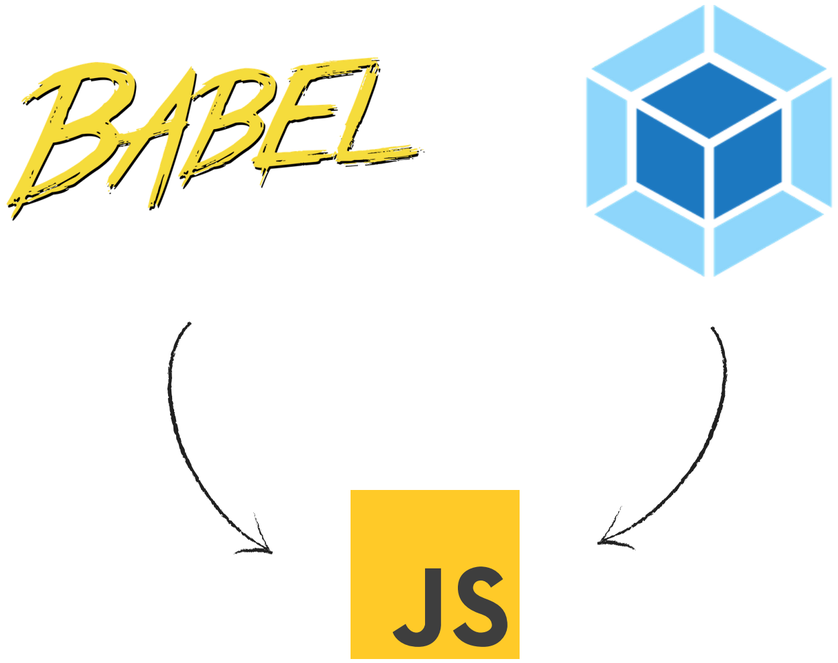
With Create-React-App Library



With Vite

Tools Behind

With Create-React-App Library



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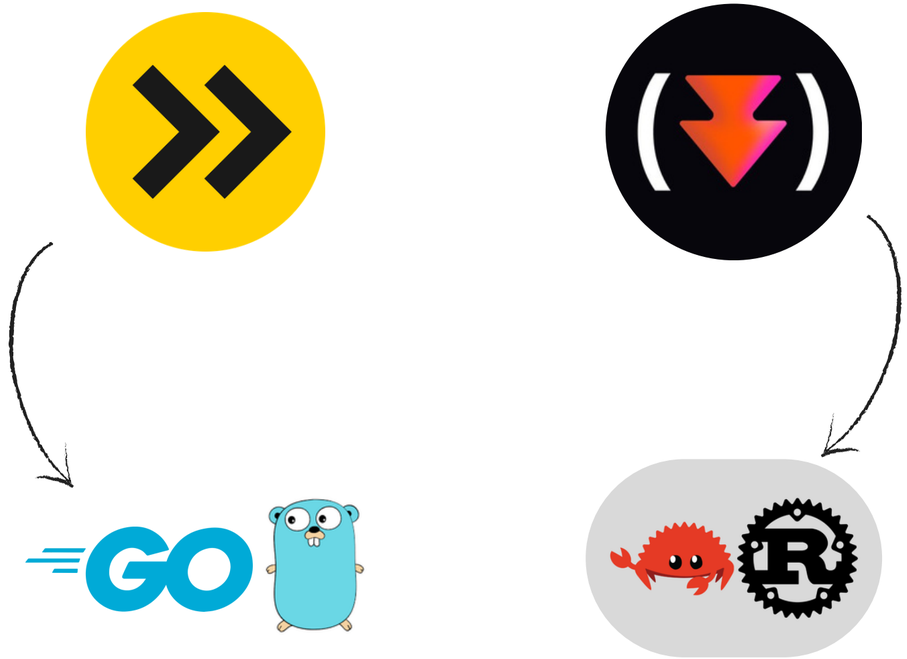


Create React App

Step 1: Add a DOM Container to the HTML

First, open the HTML page you want to edit. Add an empty `<div>` tag to mark the spot where you want to display something with React. For example:

```
<!-- ... existing HTML ... -->

<div id="like_button_container"></div>

<!-- ... existing HTML ... -->
```

Step 2: Add the Script Tags

Next, add three `<script>` tags to the HTML page right before the closing `</body>` tag:

```
  <!-- ... other HTML ... -->

  <!-- Load React. -->
  <!-- Note: when deploying, replace "development.js" with "production.min.js". -->
  <script src="https://unpkg.com/react@17/umd/react.development.js" crossorigin></script>
  <script src="https://unpkg.com/react-dom@17/umd/react-dom.development.js" crossorigin></script>

  <!-- Load our React component. -->
  <script src="like_button.js"></script>

</body>
```

```
// ... the starter code you pasted ...

const domContainer = document.querySelector('#like_button_container');
ReactDOM.render(e(LikeButton), domContainer);
```

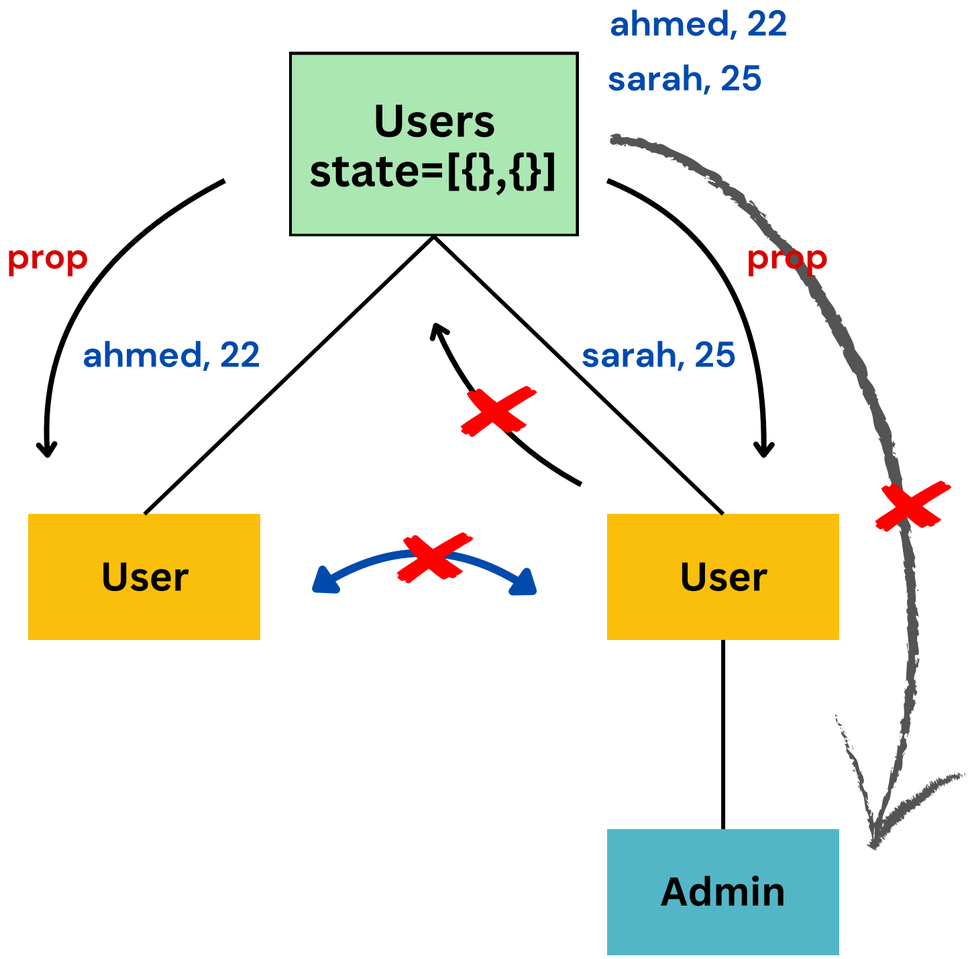
React From Scratch

Create-React-App Library

```
npx create-react-app my-app
cd my-app
npm start
```

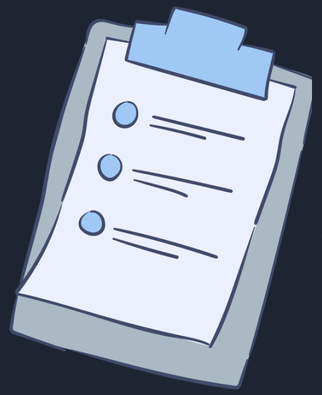
```
npm create vite@latest my-app
cd my-app
npm run dev
```

State and Props



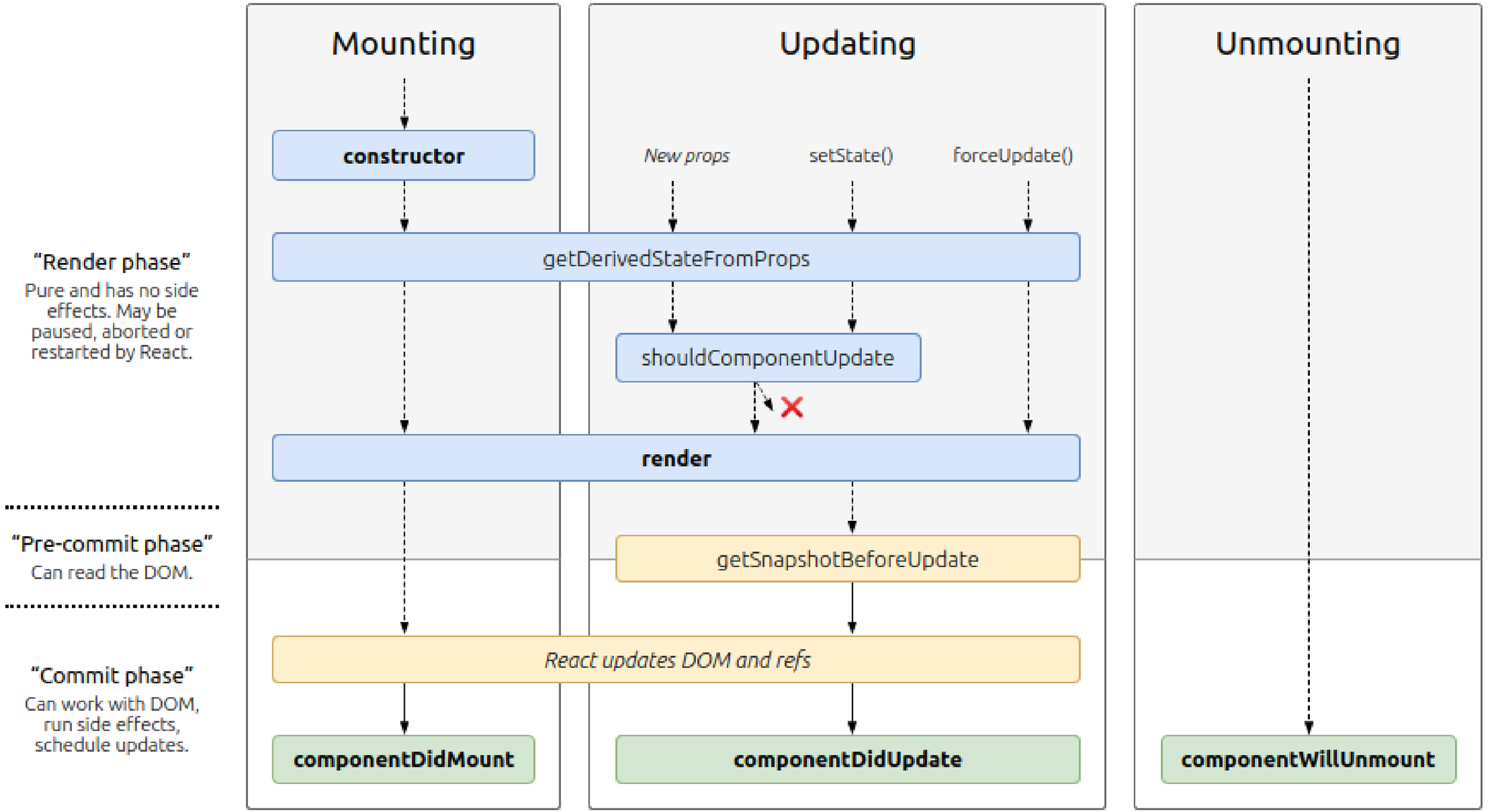
DAY 2

- Children Props
- Component Life Cycle Methods
- Events
- Call API
  - Fetch & Axios
- Dealing With Images
  - Public Folder
    - ❌ Images are not included in the build process (no optimizations)
    - ❌ You must manually ensure files exist—no Webpack/Vite validation.
  - src Folder
    - ✅ Webpack/Vite optimizes and hashes images (better Compression + Caching)
    - ✅ Ensures images exist at compile time Webpack/Vite validation

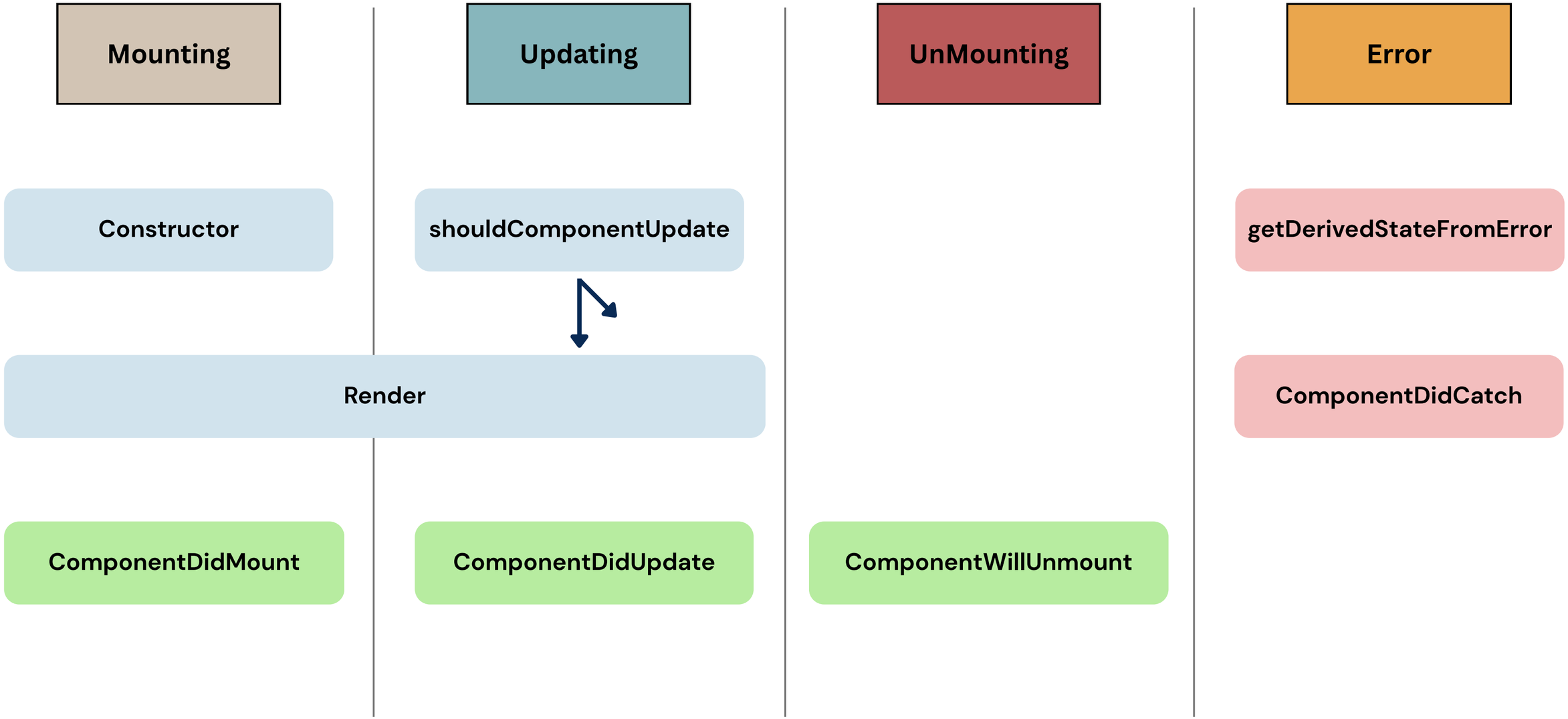


Agenda

Component Life Cycle



Component Life Cycle

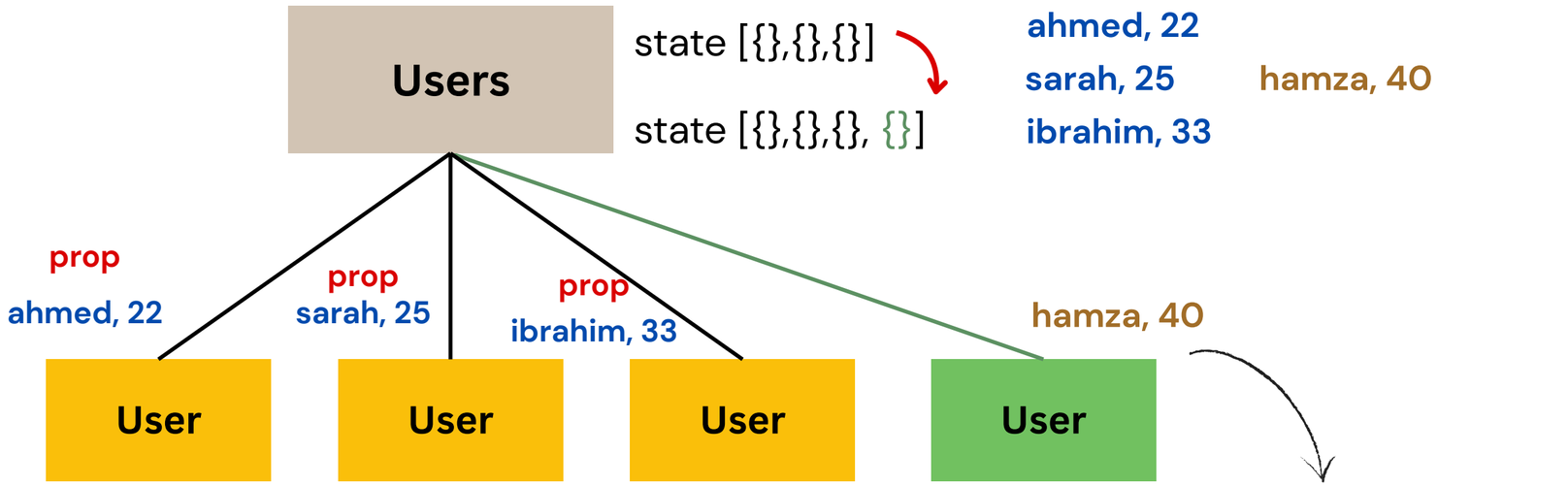


Trigger Update and Useless Rerender

Trigger Update

- state
- props
- ForceUpdate
- parent and child relationship

cause useless rerender



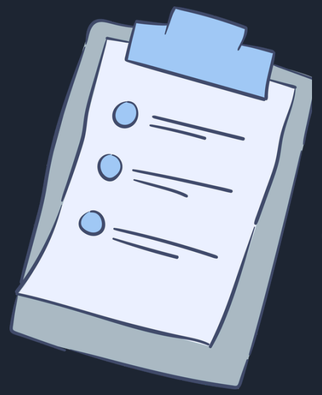
cause useless rerender

Avoid Useless Rerender

- shouldComponentUpdate(nextProps, nextState)
- PureComponent

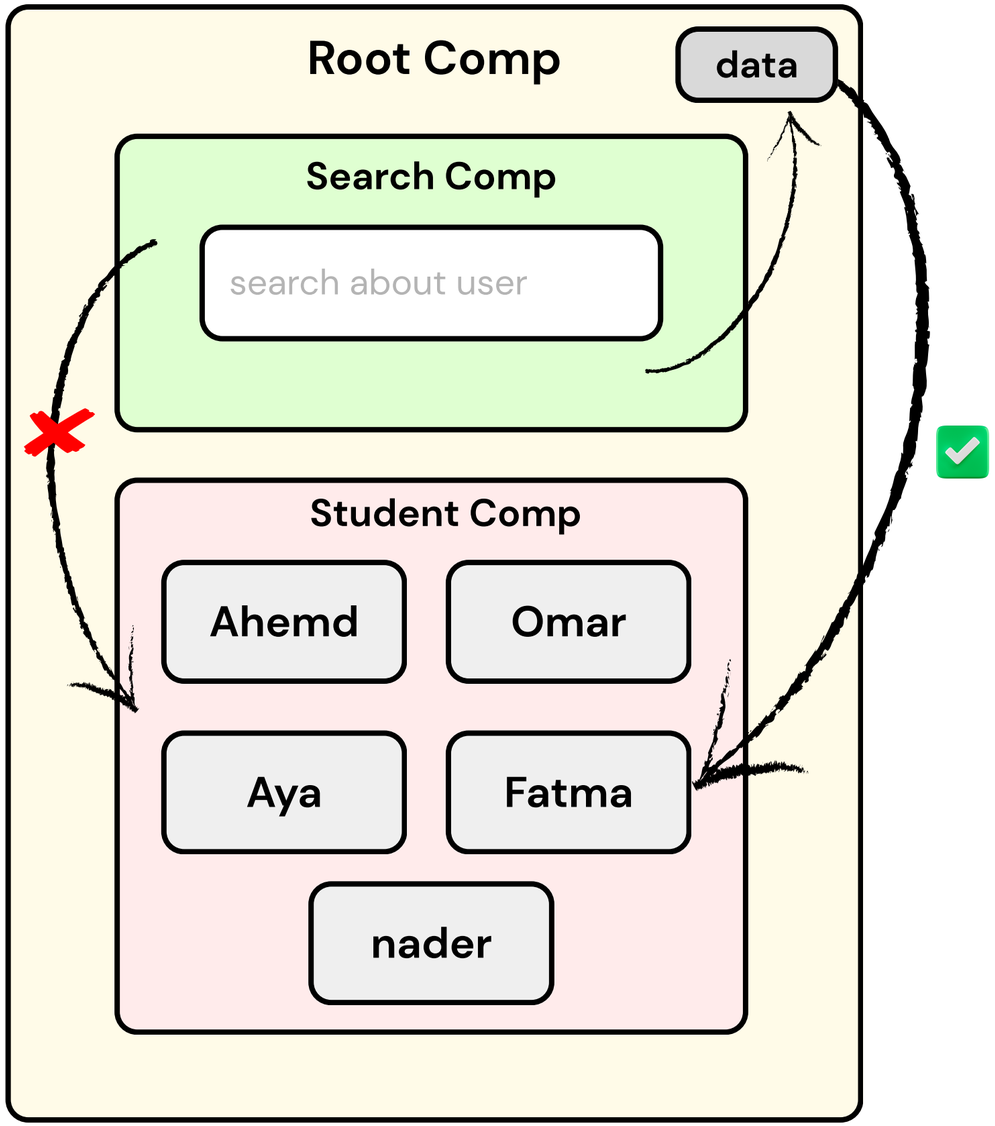
DAY 3

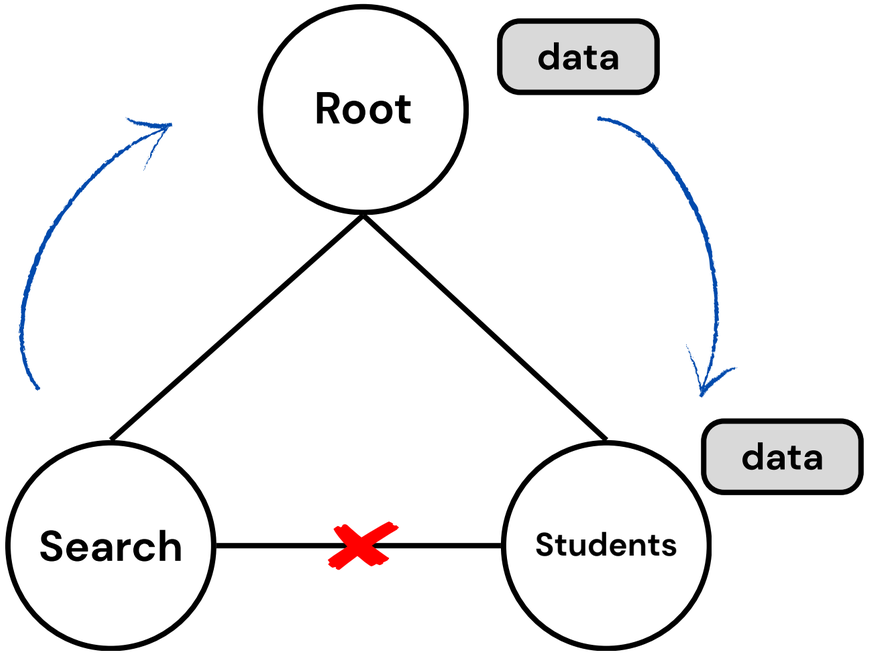
- Props Changes & Send data from child to parent
- Form
- Components Interaction
- Styling
  - Bootstrap
  - css modules



Agenda

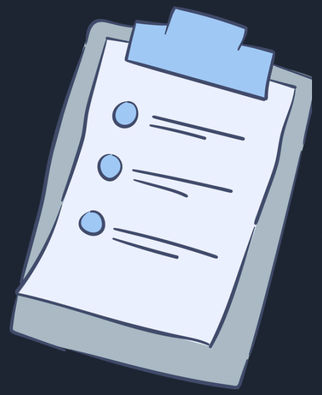
Components Interaction





DAY 4

- Intro About Hooks and history
- useState hook
  - disabled logic
- useEffect
  - intro & Take All Cases
- useCallback
- useMemo
- useRef
- useReducer
- Custom Hook



Agenda


React hooks
Introduced in React V16.8

Making Function Components its own state. Not Only that!

- Classes confuse both people and machines
- It’s hard to reuse stateful logic between components
- And more awesome powerful functionalities which we will discover some of them through our Demo
- Used in func comp or in custom hook only

THANK YOU